

Q1:

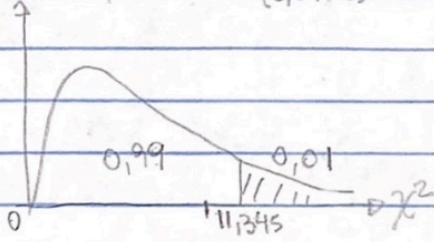
1:) $H_0: P_1 = P_2$
 $H_1: P_1 \neq P_2$

2:) $Q^2 = \sum_{i=1}^2 \sum_{j=1}^4 \frac{(O_{ij} - E_{ij})^2}{E_{ij}} \sim \chi^2_{(2-1) \times (4-1)}$
 $1 \times 3 = 3$

3:) $\alpha = 0,01$ $\chi^2_{(0,01, 3)} = 11,345$

4:) $e_{11} = \frac{58 \cdot 21}{100} = 12,18$
 $e_{12} = \frac{58 \cdot 32}{100} = 18,56$
 $e_{13} = \frac{58 \cdot 38}{100} = 22,04$
 $e_{14} = \frac{58 \cdot 9}{100} = 5,22$
 $e_{21} = \frac{42 \cdot 21}{100} = 8,82$ $e_{22} = \frac{42 \cdot 32}{100} = 13,44$ $e_{23} = \frac{42 \cdot 38}{100} = 15,96$
 $e_{24} = \frac{42 \cdot 9}{100} = 3,78$

RC: $\{Q^2 \in \mathbb{R}^+ \mid Q^2 \geq 11,345\}$



4: passo) $Q^2_{obs} = \frac{(15 - 12,18)^2}{12,18} + \frac{(22 - 18,56)^2}{18,56} + \frac{(18 - 22,04)^2}{22,04} + \frac{(3 - 5,22)^2}{5,22} +$
 $\frac{(6 - 8,82)^2}{8,82} + \frac{(10 - 13,44)^2}{13,44} + \frac{(20 - 15,96)^2}{15,96} + \frac{(6 - 3,78)^2}{3,78} = 7,0838$

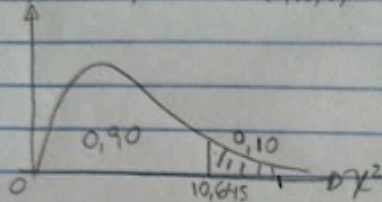
5:) $Q^2_{obs} = 7,0838 \notin RC$, portanto, não se rejeita a hipótese de igualdade das populações, ao nível de 1% de significância. Assim, não se pode afirmar que as notas obtidas por estudantes de escolas públicas ou particulares sejam diferentes.

Q2:

- 1º) $\begin{cases} H_0: \text{Renda e N.º de filhos são independentes} \\ H_1: \text{Renda e N.º de filhos não são independentes} \end{cases}$

$$2^\circ) Q^2 = \sum_{i=1}^3 \sum_{j=1}^4 \frac{(O_{ij} - E_{ij})^2}{E_{ij}} \sim \chi^2_{(3-1) \times (4-1)} = 0.2 \times 3 = 6$$

3º) $\alpha = 0,10 \quad \chi^2_{(6, 0,10)} = 10,645$



$$RC = \{Q^2 \in \mathbb{R}^+ | Q^2 > 10,645\}$$

$$4^\circ) e_{11} = \frac{135 \cdot 48}{250} = 25,92 \quad e_{12} = \frac{135 \cdot 70}{250} = 37,8 \quad e_{13} = \frac{135 \cdot 71}{250} = 38,34$$

$$e_{14} = \frac{135 \cdot 61}{250} = 32,94 \quad e_{21} = \frac{75 \cdot 48}{250} = 14,4 \quad e_{22} = \frac{75 \cdot 70}{250} = 21$$

$$e_{23} = \frac{75 \cdot 71}{250} = 21,3 \quad e_{24} = \frac{75 \cdot 61}{250} = 18,3 \quad e_{31} = \frac{40 \cdot 48}{250} = 7,68$$

$$e_{32} = \frac{40 \cdot 70}{250} = 11,2 \quad e_{33} = \frac{40 \cdot 71}{250} = 11,36 \quad e_{34} = \frac{40 \cdot 61}{250} = 9,76$$

$$Q^2_{obs} = \sum_{i=1}^3 \sum_{j=1}^4 \frac{(O_{ij} - e_{ij})^2}{e_{ij}} = \frac{(15 - 25,92)^2}{25,92} + \frac{(22 - 37,8)^2}{37,8} + \frac{(50 - 38,34)^2}{38,34} +$$

$$+ \frac{(43 - 32,94)^2}{32,94} + \frac{(25 - 14,4)^2}{14,4} + \frac{(30 - 21)^2}{21} + \frac{(12 - 21,3)^2}{21,3} + \frac{(8 - 18,3)^2}{18,3} +$$

$$+ \frac{(8 - 7,68)^2}{7,68} + \frac{(13 - 11,2)^2}{11,2} + \frac{(9 - 11,36)^2}{11,36} + \frac{(10 - 9,76)^2}{9,76} = 36,6212$$

5º) $Q^2_{obs} = 36,6212 \in RC$, assim, ao nível de 10% de significância, pode-se afirmar que as variáveis Renda, N.º de filhos não são independentes.

Q3:

$$3- \sum_{i=1}^5 x_i^2 = 3198 \quad \sum_{i=1}^5 x_i = 126 \quad \sum_{i=1}^5 y_i^2 = 30611 \quad \sum_{i=1}^5 y_i = 391 \quad \sum_{i=1}^5 x_i y_i = 9880$$

$$\bar{x} = 25,2 \quad p_x = 2,3875 \quad p_x^2 = 5,7 \quad \bar{y} = 78,2 \quad p_y = 2,9496 \quad p_y^2 = 8,7$$

$$a = 48,5789 \quad b = 1,1754 \quad r = 0,9514 \quad t_{0,05} = 5,3529$$

$$3. a) r = \frac{p_{xy}}{\sqrt{p_{xx} p_{yy}}} \quad p_{xy} = \sum_{i=1}^n x_i y_i - \frac{\sum_{i=1}^n x_i \sum_{i=1}^n y_i}{n} = 9880 - \frac{126 \cdot 391}{5}$$

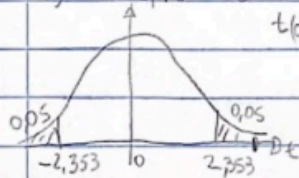
$$p_{xx} = \sum_{i=1}^n x_i^2 - \frac{(\sum_{i=1}^n x_i)^2}{n} = 3198 - \frac{126^2}{5} = 22,8$$

$$p_{yy} = \sum_{i=1}^n y_i^2 - \frac{(\sum_{i=1}^n y_i)^2}{n} = 30611 - \frac{391^2}{5} = 34,8$$

$$r = \frac{26,8}{\sqrt{22,8 \cdot 34,8}} = 0,9514 //$$

$$b) 1) \begin{cases} H_0: \rho = 0 \\ H_1: \rho \neq 0 \end{cases} \quad 2) \text{Supondo } X \text{ e } Y \rightarrow T = r \sqrt{\frac{n-2}{1-r^2}} \sim t_{n-2}$$

$$3) \alpha = 0,10 \quad \frac{\alpha}{2} = 0,05 \quad t_{(0,05;3)} = 2,353$$



$$RC = \{t \in \mathbb{R} \mid t \leq -2,353 \text{ ou } t \geq 2,353\}$$

$$4) t_{0,05} = 0,9514 \sqrt{\frac{5-2}{1-0,9514^2}} = 5,3529$$

5) $t_{0,05} \in RC$. Logo, rejeita-se H_0 ao nível de 10% de significância. Assim, as variáveis tempo e temperatura estão correlacionadas.

$$3. c) b = \frac{p_{xy}}{p_{xx}} = \frac{26,8}{22,8} = 1,1754 // \quad \bar{y} = \frac{\sum_{i=1}^n y_i}{n} = \frac{391}{5} = 78,2 //$$

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n} = \frac{126}{5} = 25,2 // \quad a = \bar{y} - b\bar{x} = 78,2 - 1,1754 \cdot 25,2$$

$$a = 48,5789 //$$

$$\hat{y} = 48,5789 + 1,1754x //$$